1. Scenario 1: Book Appointment

1.1 Fully Dressed Use Case

Use Case Name

Book Appointment

Scope

Appointment Tracking System

Level

User Goal

Primary Actor

Patient or Administrative Staff

Stakeholders

- Patient wants an available appointment.
- Healthcare Professional wants appointments within available schedule.
- Clinic wants efficient scheduling.

Preconditions

- Patient exists in the system.

- Healthcare Professional exists in the system.
- Schedule is available.

Success Scenario (Main Flow)

1. Patient or Administrative Staff requests an appointment.
2. System requests patient identification.
3. User enters patient information.
4. System retrieves patient record.
5. User selects healthcare professional and desired date/time.
6. System checks professional availability.
7. System verifies scheduling constraints.
8. System determines the time slot is available.
9. System creates appointment.
10. System marks appointment as Confirmed.
11. System displays confirmation details.

Failure Scenario

Professional Unavailable

1. User requests appointment.
2. System checks availability.
3. Selected slot is unavailable.
4. System displays available alternative slots.
5. User selects another time or cancels request.

Postconditions

- Appointment is stored in the system.

1.2 Related SSD: Book Appointment

1.3 System Operations

- searchPatient(criteria)
- selectPatient(patientId)
- checkAvailability(professionalId, locationId, serviceId, date)
- requestAppointment(patientId, professionalId, locationId, serviceId, dateTime, referralId)
- confirmAppointment(appointmentId)

1.4 Operation Contract

Contract CO1: requestAppointment

Operation: requestAppointment(patientId, professionalId, locationId, serviceId, dateTime, referralId)

Cross References: Use Case Book Appointment

Preconditions:

- Patient exists.
- Healthcare professional exists and practices at the selected clinic location.

- Selected slot available.
- A valid referral is provided if the service requires one.

Postconditions:

- Appointment instance created (instance creation).
- Appointment associated with Patient (association formed).
- Appointment associated with HealthcareProfessional (association formed).
- Appointment associated with MedicalService (association formed).
- Appointment associated with ClinicLocation (association formed).
- Appointment associated with Referral, if one was provided (association formed).
- Appointment associated with the AdministrativeStaff who made the request, if staff made it (association formed).
- Appointment status set to Requested (attribute modification).

2. Scenario 2: Register Arrival

2.1 Fully Dressed Use Case

Use Case Name

Register Arrival

Scope

Appointment Tracking System

Level

User Goal

Primary Actor

Administrative Staff

Stakeholders

- Patient wants the consultation to proceed without delay.
- Healthcare Professional wants an accurate record of which patients have arrived.
- Clinic wants reliable attendance information.

Preconditions

- A Confirmed appointment exists for the patient.

- The appointment is scheduled for today.

Success Scenario

1. Patient arrives at the clinic.
2. Administrative Staff identifies the patient's appointment for today.
3. Administrative Staff registers the patient's arrival for that appointment.
4. System verifies that the appointment exists, is Confirmed, and is scheduled for today.
5. System sets the appointment status to Arrived.
6. System confirms that the arrival has been recorded.

Failure Scenario

No Valid Appointment for Today

1. Patient arrives at the clinic.
2. Administrative Staff identifies the patient's appointment and registers the arrival.
3. System determines that the appointment does not exist, is Cancelled, or is not scheduled for today.
4. System displays the error "No valid appointment for today".
5. System rejects the arrival registration and leaves the appointment unchanged.
6. The use case ends. Administrative Staff may start Book Appointment for the patient.

Postconditions

- Success: the appointment status is Arrived.
- Failure: no appointment is changed.

2.2 Related SSD: Register Arrival

2.3 System Operations

- registerArrival(appointmentId)

2.4 Operation Contract

Contract CO2: registerArrival

Operation: registerArrival(appointmentId)

Cross References: Use Case Register Arrival

Preconditions:

- Appointment exists.
- Appointment status is Confirmed.
- Appointment is scheduled for the current day.

Postconditions:

- Appointment status changed from Confirmed to Arrived (attribute modification).

3. Scenario 3: Conduct Consultation

3.1 Fully Dressed Use Case

Use Case Name

Conduct Consultation

Scope

Appointment Tracking System

Level

User Goal

Primary Actor

Healthcare Professional

Stakeholders

- Patient wants to receive care, and wants clinical information kept confidential.
- Healthcare Professional wants complete patient history and an accurate record of the visit.
- Clinic wants accurate clinical records and accountability for access to them.

Preconditions

- The patient’s appointment exists and its status is Arrived.
- The Healthcare Professional is authorized to access clinical information.

Success Scenario

1. Healthcare Professional starts the consultation for the patient’s appointment.
2. System verifies the professional’s role, logs the access, creates the consultation, and returns the patient’s information and previous clinical records.
3. Healthcare Professional reviews the patient’s information and history.
4. Healthcare Professional conducts the consultation.
5. Healthcare Professional enters the observations, diagnosis, treatment recommendations, and follow-up requirements.
6. System creates the clinical record, linked to the consultation and the patient, and confirms that it was saved.
7. Healthcare Professional ends the consultation.
8. System sets the appointment status to Completed and displays the consultation summary.

Failure Scenario

Incomplete Clinical Record

1. Healthcare Professional submits the clinical information (step 5) with a mandatory field missing, for example the diagnosis.
2. System displays the error “Incomplete clinical record”.
3. System does not create the clinical record.

4. Healthcare Professional completes the missing information and submits again.

Postconditions

Success:

- A consultation exists, linked to the appointment and the conducting Healthcare Professional.
- A clinical record exists, linked to the consultation and the patient, holding the observations, diagnosis, treatment recommendations, and follow-up requirements.
- Any prescription or follow-up appointment requested has been created.
- The appointment status is Completed.
- The access was logged.

Failure:

- No clinical record is created.

3.2 Related SSD: Conduct Consultation

3.3 System Operations

- startConsultation(appointmentId)
- viewClinicalRecord(patientId)
- recordClinicalInfo(observation, diagnosis, treatmentRec, followUpReq)
- createPrescription()
- addMedication(medicationId, dosage, frequency, duration)

- finalizePrescription()
- requestFollowUp(dateTime, reason)
- endConsultation()

3.4 Operation Contract

Contract CO3: recordClinicalInfo

Operation: recordClinicalInfo(observation, diagnosis, treatmentRec, followUpReq)

Cross References: Use Case Conduct Consultation

Preconditions:

- Consultation has been started for the appointment.
- Healthcare Professional is authorized to access clinical information.

Postconditions:

- ClinicalRecord instance created (instance creation).
- ClinicalRecord associated with Consultation (association formed).
- ClinicalRecord associated with Patient (association formed).
- ClinicalRecord observation, diagnosis, treatmentRec, and followUpReq set (attribute modification).